

common type of flow shop scheduling problem is the permutation flow shop, in which each product must pass through all the m machines independently in the same order⁴. In complex cases, a processing task of a product may depend on another product. It is known as “no-wait” flow shop scheduling problem when there is no delay allowed between two successive tasks of a product. Many hybrid flow shop models have been developed to reflect the reality, which is mostly specific to a production system¹⁶. To make production systems energy-efficient and environmentally friendly, many flow shop models have been proposed, which are widely known as “green flow shop model”. Here, in order to establish an efficient resource allocation, total energy consumption is taken into account in addition to makespan¹⁷.

Although modern industries have been applying many powerful decision-making tools, such as intelligent manufacturing systems, to address complex challenges^{18,19}, small and medium-sized bakeries continue to rely on personal experience^{6,7}. Furthermore, bakeries’ product range and amount change frequently due to market demand and seasonality, demanding continuous monitoring of production efficiency. However, only a few studies have focused on improving bakery manufacturing. In a recent study, Huber and Stuckenschmidt²⁰ implemented machine learning approaches to predict hourly sales of bakery items in a retailer store and optimize the baking schedule so that bakers serve customers with fresh products. Nonetheless, the production of a vast number of products, from flour to finished or unfinished goods before delivery to retailers, is a separate segment. In a case study with a medium-sized German bakery, Hecker et al.⁶ observed that the makespan of an existing manufacturing line with 40 products can be lowered by 8.6%. Swangnop et al.⁷ developed a scheduling model for a bakery in Thailand and demonstrated that the existing production, planned based on experience, is inefficient. An additional factor that most bakers overlook is the ovens’ energy consumption, which has a vital influence on manufacturing costs and CO₂ emissions. It has been reported that only baking consumes up to 78% of the total energy depending on the product category²¹. Bakeries typically feature multiple ovens with varied functionalities that are employed according to product specifications^{5,6}. As a result, by minimizing the idle time of the ovens, a large quantity of energy can be saved, lowering manufacturing costs and CO₂ emissions. Babor et al.⁵ investigated a small Spanish bakery and observed that actual production is poorly optimized. The authors weighed the machines’ idle time and makespan to the objective function.

The motivation for this study is twofold. First, to solve production optimization problems for bakeries with two objectives: minimizing makespan and energy waste due to oven idle time. It is a hybrid no-wait flow shop scheduling problem because of the following exceptions in bakery manufacturing. Many tasks are carried out manually, and there are numerous substitute machines that can carry out the remaining tasks. Additionally, there are production constraints for a variety of products recipes. Therefore, a mixed-integer linear programming approach for hybrid no-wait flow shop scheduling model (HNFSM) is proposed to simulate bakery production scheduling. The Pareto optimal solutions obtained by multi-objective optimization algorithms are used to analyze the trade-offs between the objectives. Secondly, to compare the performance of five multi-objective optimization algorithms to solve the instances. Because production optimization is time-consuming and performed frequently, attaining optimal solutions in the lowest computation time is essential. We used multi-objective optimization algorithms of two types: evolutionary algorithms and particle swarm optimization-based metaheuristics. Non-dominated sorting genetic algorithm (NSGA-II), strength Pareto evolutionary algorithm (SPEA2), and generalized differential evolution (GDE3) are taken from the former category, while improved multi-objective particle swarm optimization (OMOPSO), and speed-constrained multi-objective particle swarm optimization (SMPSO) are from the latter. To assess their effectiveness, four quality indicators are used: cardinality, convergence, distribution and spread, and convergence and distribution of the obtained solutions. To cluster the solutions into distinct qualities, a Gaussian mixture model²² is used.

The following are the contributions of the current study. State-of-the-art multi-objective optimization methods are used to optimize the production efficiency of small and medium-sized bakeries employing a hybrid no-wait flow shop model. By combining various performance metrics, the Gaussian mixture model is used to assess how effectively algorithms solve problems of three complexity levels while varying the number of products and predecessor constraints.

The remainder of the paper is structured as follows. An introduction to bakery production is given in the next section. “[Materials and methods](#)” section describes mathematical formulation of a hybrid no-wait flow shop scheduling model to simulate bakery manufacturing. Besides, multi-objective optimization algorithms and their performance indicators are presented. In “[Results and discussion](#)” section, the effectiveness of algorithms in solving scheduling problems for bakeries is analysed. “[Conclusions and future works](#)” section of this paper provides a summary of findings.

Bakery production

A bakery product undergoes a series of processing steps. Each product has a recipe that specifies the order, duration, and machines that will be utilized to complete the tasks. Figure 1 shows a simplified processing route. Making the dough starts by mixing and kneading the ingredients, such as flour, water, and salt. In most cases, yeast is added to induce fermentation, which produces the leavening agent CO₂ and other aroma precursors. If small and medium-sized bakeries are considered, the transfer of unfinished products from one machine to another is performed manually by employees almost after every processing task. The fermentation process has a vital impact on the final product’s quality. Because temperature and humidity have a considerable impact on yeast sugar fermentation, the duration of the processing stages under various conditions is strictly controlled. Performing one task longer than the predefined duration results in overtreatment and consequently, delay for the following tasks, and a loss of product quality, both of which are undesirable. Therefore, as soon as ingredients are mixed and kneaded, the next stages are carried out with no delay.

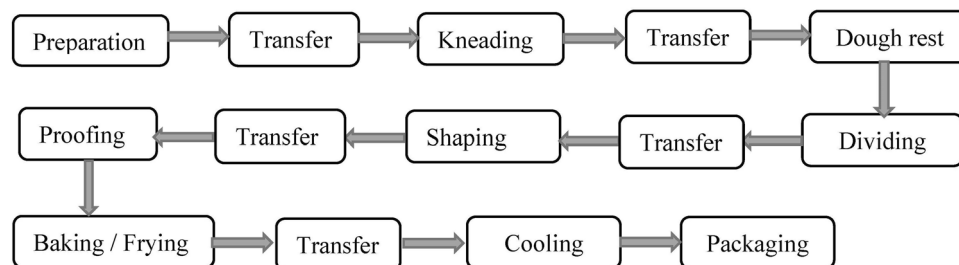


Figure 1. A simplified processing route for bakery products.

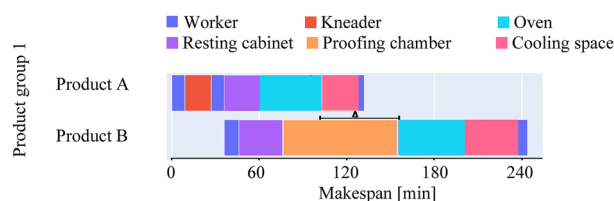


Figure 2. Gantt chart showing schedule of one product group with unified initial stages. The triangle (Δ) shows the oven idle time.

To avoid wasting energy, bakers turn off machines during the idle time—the time between two scheduled operations. However, ovens need preparation time to reach the set temperature before performing baking. Turning them off after an operation requires a restart well ahead of the next operation. When they are turned off, no energy is consumed, but the temperature steadily declines. If the idle time between two tasks is short, the temperature drops less, and the time required to re-heat is reduced. However, if the temperature drops sharply, such as due to chilly weather, the assumed time may not be long enough to reach the set point. Again, with the prolonged idle period, the right time to restart the ovens must be considered to avoid wasting energy and have them ready at the proper time. When the number of products is large and there are many manual tasks to perform, it is difficult for bakers to keep track. The following tasks must be postponed accordingly if the oven's temperature is not up to the set point in time. It may affect the product quality and lead to inefficient production. To avoid these consequences, bakers keep the ovens running throughout the production time.

The duration and machine set up for processing steps are predetermined. Hence, the energy consumption during operational time is constant regardless of how optimized a schedule is. In contrast, the effectiveness of the schedule influences oven idle time, which has a direct impact on the quantity of wasted energy. Moreover, small, medium, and medium-sized manufacturers have limitations in recording energy data for each device. In this case, the idle time of the ovens can be an ideal indicator of energy waste. Machine idle time has been investigated in various studies as one of the objectives for optimizing production schedule^{5,6,23,24}.

Materials and methods

In this paper, three bakery production optimization problems, labelled with BK15, Bk40 and BK50 were solved. The number in labels specifies how many products were produced in each manufacturing line, for example, the dataset BK15 contains production information for 15 bakery products. BK15 and BK50 were taken from Babor and Hitzmann²⁵ and BK40 was taken from Hecker et al.⁶. BK15 had three employees, eight machines, and two ovens with four compartments; BK40 had eleven machines, three ovens, and nine compartments; and BK50 had ten employees, fourteen machines, and three ovens, and ten compartments. In bakeries, many ovens have separate compartments, each of which can be used independently to bake a batch of products. In the following discussion, the problems are labelled according to the approximate number of total products. The implementation and simulation of HNFSM and multi-objective optimization algorithms were performed using the computer language Python (version 3.7)²⁶ on a computer running Microsoft Windows 10 as the operating system with a configuration of an Intel Core i5 at 4 × 3.20 GHz, 8 GB ram.

Problem definition and scheduling model. In small, and medium-sized bakeries, using the same dough for various products made from the same ingredients is a frequent practice. This practice takes advantage of the machines' capabilities in the initial stages to reduce preparation time. Bakers split the dough after completing a few processing tasks into various parts. It enables the products to be treated differently in subsequent phases to meet recipe requirements. There is no common procedure for separating dough as it completely depends on the type of products and recipes. Figure 2 illustrates a schedule of two products that are produced from the same dough and shared the same processing machines at the initial phase.

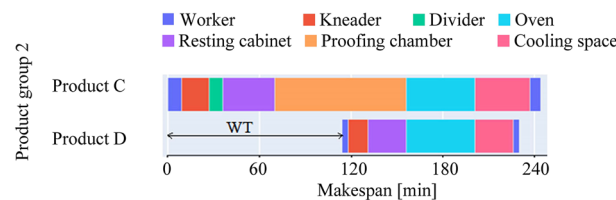


Figure 3. Simplified schedule for one product group where baking is performed together. Here WT is the difference between the start time of Product C and Product D.

Group	Product	Bowl time [min]	Name	Processing stage	Duration [min]	Machine/Employee
1	1	0	Product A	Preparation	9	Employee
				Kneading	18	Kneader
				Dividing	9	Employee
				Dough rest	25	Resting cabinet
				Baking	42	Oven
				Cooling	25	Cooling space
				Packaging	4	Employee
	2	36	Product B	Dividing	10	Employee
				Dough rest	30	Resting cabinet
				Proofing	80	Proofing chamber
				Baking	45	Oven
				Cooling	36	Cooling space
				Packaging	7	Employee

Table 1. Simplified production data for one product group.

In another scenario, multiple products that came across different processing routes are baked in the same oven. Because baking consumes high energy, running an oven while it is only partially occupied causes energy waste. Figure 3 shows a schedule for two products where baking is performed together. In both cases (Figs. 2, 3), the products are internally dependent such that their common tasks must be performed at the same time. This preceding rule is used to arrange products into groups in the flow shop model. Only one product in a group has no predecessor, which means it can be scheduled at any time throughout the production runtime. However, the schedules for the rest of the products in that group depend on it. Table 1 represents simplified production data for one product group which is visualized in Fig. 2.

In reality, many product groups are organized based on their internal dependence. Within a group, each product has an individual bowl time. It indicates the start time difference between a predecessor product and any other product in a group. For one processing stage, there might be multiple alternative machines and employees, from which one should be selected based on availability. In general, for dough rest and cooling, no energy-consuming machines are required and therefore are considered to have the capacity to operate as many products as possible at a time. Similarly, due to having enough space in proofing chambers, it is assumed that the proofing stage has no blockage. Considering the bakers' practice, ovens can operate multiple products from the same product group at a time (Fig. 3). The rest of the machines can perform a task only from one product.

Figure 4 shows the procedure of optimizing the bakery production schedule. It can be discussed in three distinct segments: data collection, HNFSM, and optimization algorithm. Information about bakery products, machines, and employees is recorded during data collection. Depending on the internal dependence the products are sorted into distinct groups. An initial product sequence, an order of product groups in which they are produced, is transferred to HNFSM. In HNFSM, the processing tasks are allocated among the machines and employees. Here, the actual scheduling, i.e., exact start and end time, machine, or employee to conduct a task is determined. Like other flow shop models, the products that are placed first in the order will get priority in occupying the machines and employees. The following products are scheduled according to machine availability. The makespan and OI DT are calculated as a quality indicator for a schedule, which is considered as a baseline to start improving. The optimization algorithm proposes a new candidate solution vector, which requires to convert into a product group sequence to use for HNFSM. The candidate solution conversion approach is explained later. This procedure is repeated until a certain termination criterion is met.

This section describes the mathematical formulation of the proposed hybrid no-wait flow shop scheduling model. Table 2 defines the notions that are used to describe HNFSM. Ovens at bakeries typically contain multiple compartments that may each be used separately to bake different products. OI DT may exceed the makespan since it calculates the total idle time of all oven compartments. The following equations are used to calculate makespan and OI DT.